List of Symbols

γ stoichiometric coefficient.

δ death rate.

M microbe biomass.

N_P number of pathways.

N_S system size, number of substance/microbe pools.

R gas constant.

S_i chemical/microbe species.

T, T° sample temperature, standard temperature.

i index over chemical/microbe species.

j index over pathways.

t time.

v turnover rate.

$v_{\max}$ maximum pathway turnover velocity.

1 Methods

1.1 Pathways

The model takes into account N_P pathways, each characterized by a reaction equation and associated with a group of microbes. Each pathway is characterized by a reaction equation.

1.1.1 Aceto

$$\mathrm{CH_3COOH} \rightleftharpoons \mathrm{CO_2} + \mathrm{CH_4}$$

1.1.2 Iron reduction

$$\mathrm{CH_3COOH} + \mathrm{H_2O} + 8\,\mathrm{Fe^{3+}} \rightleftharpoons 8\,\mathrm{Fe^{2+}} + 2\,\mathrm{CO_2}$$

1.2 System

The system is composed of several substances and microbe populations, both changing over time. The amount of the i-th substance (chemical or microbial species) that is present in the system at time t is

$$S_i = S_i(t). \tag{1}$$

The amount of biomass for the microbial pools are given by

$$M_j = M_j(t). \tag{2}$$

The entire system state is described by the array

$$\boldsymbol{S} = \begin{bmatrix} S_0 \\ S_1 \\ \dots S_{N_S} \\ M_0 \\ M_1 \\ \dots \\ M_{N_P} \end{bmatrix} \tag{3}$$

The microbes' metabolism is composed of an anabolistic and a catabolistic fraction. The reaction equation describes the catabolism. In addition, the microbes use one substance as carbon source for the anabolism. The carbon-use-efficiency CUE_j indicates the fraction of carbon turnover which goes into growth. Hence, the carbon turnover of the total metabolism is composed of the stoichiometric coefficient for the respective carbon source and additional terms describing the anabolism, i.e.

$$\mathrm{CUE}_j = \frac{A_j}{A_j + |\Gamma_j|}, \tag{4}$$

where $|\Gamma_j|$ is the stoichiometric coefficient of the substance used as carbon source by the microbes of the j-th pathway. Rearranging yields the effective stoichiometric contribution of the anabolism for the j-th pathway

$$\alpha_j = |\Gamma_j| \frac{\mathrm{CUE}_j}{1 - \mathrm{CUE}_j}. \tag{5}$$

This term describes the reduction of substance used as carbon source for the anabolism by the j-th pathway's microbes.

The evolution of the system is described via the rates of change

$$\frac{\mathrm{d}S_i}{\mathrm{d}t} = \sum_{j=1}^{N_P} M_j v_j \left(\gamma_{ij} - \alpha_{ij}\right), \tag{6}$$

$$\frac{\mathrm{d}M_j}{\mathrm{d}t} = \sum_{j=1}^{N_P} M_j v_j \left(\alpha_{ij} - \delta\right) \tag{7}$$

for the substances and microbes, respectively. The stoichiometric coefficients for the i-th substance, γ_{ij}, are negative for reactants and positive for products. Furthermore,

$$\alpha_{ij} = \begin{cases} A_j & \text{if } i = \text{carbon source index} \\ 0 & \text{otherwise,} \end{cases} \tag{8}$$

describes the anabolism, transferring carbon from the source to the driving microbe group of the j-th pathway. The diminishing rate δ is constant for all microbe groups.

The j-th pathway's turnover rate is governed by the velocity

$$v_j = v_{\max,j} \cdot f_{\mathrm{MM},j} \cdot f_{\mathrm{thermo},j} \cdot f_{\mathrm{inhib},j}, \tag{9}$$

with the constant maximum turnover rate $v_{\max}$ and the time-dependent dimensionless factors $f_{\cdot,j}$ with values between 0 and 1, summarizing the effects of kinetics (Michaelis-Menten and inhibition) and thermodynamics at time t.

The forward and inverse kinetics are modelled by the factors

$$f_{\mathrm{MM},j} = \prod_{i=1}^{N_\mathrm{S}} \frac{S_i}{S_i + K_{m,ij}} \tag{10}$$

and

$$f_{\mathrm{inhib},j} = \prod_{i=1}^{N_\mathrm{S}} \frac{S_i}{S_i + K_{\mathrm{inhib},ij}}, \tag{11}$$

respectively, where $K_{m,ij}$ and $K_{\mathrm{inhib},ij}$ are the half-saturation coefficients with respect to the i-th substance in the j-th pathway. If the i-th pool is irrelevant for the j-th pathway, $K_{m,ij} = 0$. Numerics: Add $\varepsilon = 10^{-12}$ in the denominator, to avoid division by zero.

The thermodynamics is taken into account by the factor

$$f_{\mathrm{thermo},j} = 1 - e^{\frac{\min(0, \Delta G_{r,j} - \Delta G_{r,\min})}{RT}}, \tag{12}$$

with the energy

$$\Delta G_{r,j} = RT \log Q_j + \sum_{i=1}^{N_P} \gamma_{ij} \Delta G_{f,i} \tag{13}$$

defined via the reaction quotient of the species' activities a_i for each pathway,

$$Q_j = \prod_{i=1}^{N_P} a_i^{\gamma_{ij}}, \tag{14}$$

and the constant formation energies $\Delta G_{f,i}$. The Gibbs minimum $\Delta G_{r,\min} = -26 \cdot 10^3 _\frac{\mathrm{kJ}}{\mathrm{mol}}$ defines the boundary of thermodynamic viability.

The system behaviour is parametrised by a set of model parameters $\boldsymbol{\theta}$

$$\frac{\mathrm{d}\boldsymbol{S}}{\mathrm{d}t} = f(t, \boldsymbol{S}; \boldsymbol{\theta}) \tag{15}$$

and can be integrated numerically using an initial system state $\boldsymbol{S}_0$ to yield $\boldsymbol{S}(t)$.

$$\boldsymbol{x} = \begin{bmatrix} \boldsymbol{\theta} \\ S_{\mathrm{Fe},0} \\ S_{\mathrm{Ac},0} \\ M_{\mathrm{j},0} \end{bmatrix} \tag{16}$$

$$\boldsymbol{\theta} = \begin{pmatrix} K_{m,\mathrm{Ferm,Hydrolysis}} \\ K_{m,\mathrm{DOC,Ferm}} \\ K_{m,\mathrm{H_2,Hydro}} \\ K_{m,\mathrm{CO_2,Hydro}} \\ K_{m,\mathrm{H_2,Homo}} \\ K_{m,\mathrm{CO_2,Homo}} \\ K_{m,\mathrm{Ac,Aceto}} \\ K_{m,\mathrm{Ac,Fe^{3+}}} \\ K_{m,\mathrm{Fe^{3+},Fe^{3+}}} \\ K_{m,ij} \\ K_{\mathrm{inhib},ij} \\ \mathrm{CUE}_j \end{pmatrix} \tag{17}$$

2 Fit

The model has a number of parameters that need to be tuned using the incubation data.

The incubation data provides measurements of total CO_2 and CH_4 released from the soil at time t_k.

For a given set of model parameters $\boldsymbol{x}$,

A Thermodynamics

$$H_{cc,i} = H_{cp}^{\circ} R T e^{C_{cp}\left(\frac{1}{T} - \frac{1}{T^{\circ}}\right)} \tag{18}$$

$$\lambda_i = H_{cc,i} \frac{V_{\mathrm{aq}}}{V_{\mathrm{gas}}} \tag{19}$$

$$S_i^{\mathrm{dis}} = S_i \frac{\lambda_i}{1 + \lambda_i} \tag{20}$$

$$a_i = \frac{S_i^{\mathrm{dis}}}{w} \tag{21}$$

$$\Delta G_{s,j} = \sum_{i=1}^{N_P} \gamma_{ij} \Delta G_{f,i} \tag{22}$$

In order to avoid negative pools,

$$S_i = \max(0, S_i) \tag{23}$$